

A Comparative Evaluation of Context Engineering Approaches for the Modernization of Legacy Enterprise Web Applications

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Chapter 1

Introduction

1.1 Research Topic and Context

This section will introduce the broad research topic and the context of why identifying reliable AI assisted approaches to modernizing legacy software applications is valuable.

1.2 Project Objectives

This section will describe the gap in research that the study will address and the contribution it will make to the literature. It will also outline the goals and objectives of the study.

1.3 Research Questions

This section will outline the research questions that the study aims to answer.

1.4 Outline of Chapters

This section will provide an outline of the chapters in the dissertation including a brief sequential description of each.

1.5 Datasets Used

This section will describe the benchmark repositories that will be used. It will also provide details of any similar studies that has been performed on the benchmark repositories.

Chapter 2

Literature Review

2.1 Legacy Application Modernization

2.1.1 Legacy Application and Technical Debt

This section will provide context about the problems legacy applications and technical debt can cause enterprises. This may include ongoing risks legacy applications pose from a security and customer satisfaction perspective.

2.1.2 Factors Driving Application Modernization

This section will outline the factors that drive the need to modernize enterprise web applications. This could include evolving web application frameworks, end of life support from software vendors or evolving business needs.

2.1.3 Application Modernization Challenges

This section will cover challenges organisations face when considering digital remediation and modernisation projects from a resourcing perspective.

2.1.4 Scaling to Enterprise Legacy Systems

This section will consider the complexity of enterprise web applications and the challenges that an AI assisted migration approach will face in large code bases.

2.2 Large Language Models

This section will outline the core technology and concepts underlying Large Language Models (LLMs). LLMs are fundamental to context engineering so the study will need to demonstrate a clear awareness and understanding of this core non-deterministic technology.

2.2.1 Transformer Architecture

This section will outline the key concepts in the transformer architecture.

2.2.2 LLM Training

This section will outline the phases of training LLMs. It will include foundation model training, fine tuning and prompt engineering techniques.

2.2.3 Tokenization and Context Windows

This section will describe how language is translated into tokens and provided to the LLM in a context window. It will also explore limitations and increases of the context window size over time.

2.2.4 Retrieval Augmented Generation

This section will introduce the concept of Retrieval Augmented Generation (RAG) and how it is used to provide additional context information to the LLM. It will describe the process of creating vector embeddings to represent this information, vector embedding databases and the algorithms used for searching and ranking information to be provided to the LLM.

2.2.5 Prompt Engineering

This section will describe approaches to prompt engineering and how it is used to provide additional context to LLMs. It will cover techniques such as zero shot and few shot prompting and explore best practice for structuring prompts for software engineering tasks.

2.2.6 Limitations, Reliability and Hallucinations

This section outlines the core problem with using non-deterministic LLMs to write software engineering applications which are traditionally expected to behave in a reliable, deterministic way. It will describe what is meant by hallucinations including how and why they occur as well as exploring techniques documented in the literature to mitigate this issue.

2.3 Agentic Artificial Intelligence

The dissertation topic will rely heavily on the use of AI agent platforms. This section will explore the key concepts and developments for agent architectures from the literature.

2.3.1 Agent Architectures

This section will contain an outline of the key developments in AI agent architecture including techniques such as ReAct, Chain of Thought, Reflexion etc. This may break out into further sub-sections as additional reading is completed.

2.3.2 Tool Usage

This section will describe how agentic frameworks can start to complete tasks beyond returning responses to prompts. It will explore how tools are made available to LLMs and how this applies to software engineering tasks.

2.3.3 Agentic Memory

This section will explore how agentic architectures remember context over long running workflows.

2.3.4 AI Software Engineering Agents

This section will document and describe the key proprietary AI Software Engineering agents that exist in the industry at present. It will describe the characteristics of each, categorize them and outline how they will be used in the study.

2.3.5 Agent Skills

This section will introduce the concept of AI agent skills as text files which provide additional context to the LLM with reference to the literature. It will describe the different proprietary frameworks used to implement and structure AI skills and how these will be used in the study.

2.4 Context Engineering

Context Engineering approaches are the central topic of the study. This section outlines the set of established approaches to the discipline currently practiced in industry.

2.4.1 Multi-Agent Workflows

This section will describe the concept of multi-agent workflows in context engineering with reference to the literature.

2.4.2 BMAD

This section will introduce and describe the BMAD context engineering framework as an example of a multi-agent, multi-step orchestration workflow.

2.4.3 Specification Driven Approaches

This section will describe specification driven approaches to context engineering with reference to the literature.

2.4.4 GitHub SpecKit

This section will introduce and describe GitHub SpecKit as an example of a specification driven approach and how it will be used in the study.

2.4.5 OpenSpec

This section will introduce and describe OpenSpec as another example of specification driven context engineering and how it will be used in the study.

2.5 Software Application Migration Benchmarks

2.5.1 Available Benchmarks

This section will outline the benchmark repositories of legacy application code that were explored for use in the study.

2.5.2 Selection Methodology

This section will describe the methodology and criteria used for selecting a benchmark repository for use in the study.

2.5.3 Selected Benchmark

This section will identify the selected benchmark repositories and explain why it was selected based on the selection methodology.

2.6 Code Migration Approaches

This section will survey the main approaches to the migration of software engineering applications with reference to the literature and industry practice. The following sub-headings may change as further reading is completed on this topic.

2.6.1 Manual Code Migration

This section outlines manual approaches to modernizing software engineering application.

2.6.2 Semi Automated Code Migration

This section outlines semi-automated approaches to migrating legacy code, assisted by manual effort.

2.6.3 Automated Code Migration

This section will outline automated approaches to modernizing software applications including AI assisted approaches.

2.7 Evaluation and Metrics

2.7.1 Quantitative Code Quality Metrics

This section will identity appropriate quantitative metrics to use to evaluate the output of context engineering approaches.

2.7.2 Qualitative Code Quality Metrics

This section will identity appropriate qualitative metrics to use to evaluate the output of context engineering approaches.

2.7.3 Evaluation of LLM Outputs

This section will outline and describe the use of evaluations (evals) to validate the output of LLMs with the goal of identifying how to incorporate evaluation techniques in measuring the quality of the migrated software code.

2.8 State of the Art

The goal of the study is to identify a reliable way to migrate legacy enterprise software applications to modernized implementations. This section will survey approaches to this problem that have been documented in the literature. The sub-heading in this section will expand as more reading is completed. It will identify previous studies of using AI-assisted approaches for the modernization of legacy code applications and compare results.